Hao (Mark) Chen

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RESEARCH INTERESTS

My research focuses on bridging the gap between advanced AI algorithms and hardware platforms through efficient hardware-aware algorithms, cost-effective model merging, ML-assisted hardware design, and algorithm-system co-design. My current research directions include:

- Test-time Optimization for LLMs.
- LLM quantization for edge deployment.
- Model merging.
- Code agents for hardware design.

EDUCATION

Imperial College London
PhD in Computing *Sep 2024 – June 2028 (expected)*
Advisor: Prof. Wayne Luk, Dr. Hongxiang Fan

Imperial College London
MEng in Computing *Oct 2020 – June 2024*
Advisor: Prof. Wayne Luk, Dr. Hongxiang Fan

PUBLICATIONS & MANUSCRIPTS

Journal Papers

- **Hao (Mark) Chen**, Liam Castelli, Martin Ferianc, Shuanglong Liu, Wayne Luk, Hongxiang Fan. Algorithm and Hardware Co-Design for Multi-Exit Dropout-based Bayesian Neural Networks.

Conference Papers

- **Hao (Mark) Chen**, Shell Xu Hu, Wayne Luk, Timothy Hospedales, Hongxiang Fan. FW-Merging: Scaling Model Merging with Frank-Wolfe Optimization.
- **Hao (Mark) Chen**, Zehuan Zhang, Wanru Zhao, Nicholas Lane, Wayne Luk, Hongxiang Fan. Advancing AI-assisted Hardware Design with Hierarchical Decentralized Training and Personalized Inference-Time Optimization.
- Charlie Campbell, **Hao (Mark) Chen**, Wayne Luk and Hongxiang Fan. Advancing Multi-Agent Code Generation for Quantum Programs with Iterative Semantic Optimization and Quantum Error Correction. Accepted at *2025 Design Automation Conference (DAC)*.
- **Hao (Mark) Chen**, Fuwen Tan, Alexandros Kouris, Royson Lee, Hongxiang Fan, Stylianos I. Venieris. Progressive Mixed-Precision Decoding for Efficient LLM Inference. Accepted at *2025 International Conference on Learning Representations (ICLR)*.
- Jiahao Gai, **Hao Chen**, Zhican Wang, Hongyu Zhou, Wanru Zhao, Nicholas Lane, Hongxiang Fan. Exploring Code Language Models for Automated HLS-based Hardware Generation: Dataset, Infrastructure and Analysis. *2025 Asia and South Pacific Design Automation Conference (ASP-DAC)*.
- **Hao (Mark) Chen**, Wayne Luk, Ka Fai Cedric Yiu, Rui Li, Konstantin Mishchenko, Stylianos I. Venieris, Hongxiang Fan. Hardware-Aware Parallel Prompt Decoding for Memory-Efficient Acceleration of LLM Inference.
- Zehuan Zhang, Hongxiang Fan, **Hao (Mark) Chen**, Lukasz Dudziak and Wayne Luk. Hardware-Aware Neural Dropout Search for Reliable Uncertainty Prediction on FPGA. *2024 Design Automation Conference (DAC)*.
- Hongxiang Fan, **Hao (Mark) Chen**, Liam Castelli, Zhiqiang Que, He Li, Kenneth Long, Wayne Luk. When Monte-Carlo Dropout Meets Multi-Exit: Optimizing Bayesian Neural Networks on FPGA. *2023 Design Automation Conference (DAC)*.
- **Hao (Mark) Chen**, Taowen Liu, Songyun Hu, Leyang Yu, Yiqi Li, Sihan Tao, Jacqueline Lee, Ahmed E. Fetit. Web-based AI System for Medical Image Segmentation. *2023 Conference on Medical Image Understanding and Analysis (MIUA)*.

INDUSTRIAL EXPERIENCE

Samsung AI Center
Research Intern, Cambridge, UK *July 2024 – December 2024*
Worked on efficient LLM inference involving quantization.

Qube RT
Quantative Technologist Intern, UK *April 2023 – Sep 2023*
Developed C++ performance monitoring system for low-latency trading platform.

Huawei Technologies Research & Development
Graphics Modelling Intern, UK *March 2022 – Sep 2022*
Worked on automatic GPU API generation.

Ampere Computing Shanghai, China
Java Software Developer *June 2021 – Sep 2021*
Developed a log search engine for Jenkins based on Lucene Search.

AWARDS

- **President's PhD scholarships** 2024 – 2028
- **Winton Capital Applied Undergraduate Project Computing Prize** 2024
- **Governors' MEng Prize in Computing** (the best overall performance in MEng programme) 2024
- **Dean's List (Year 3)**, Imperial College London 2023
- **Dean's List (Year 2)**, Imperial College London 2022
- **Dean's List (Year 1)**, Imperial College London 2021
- **G-Research Ltd Prize**, G-Research Ltd, 2021
- **Singapore SM1 Scholarship**, Ministry of Education Singapore 2015 – 2019

SKILLS

- **Programming:** C++, C, Python, Scala, Java, Swift, Haskell, Bash
- **Tools:** GCC, Jenkins, Github, Docker, Heroku, AWS
- **Framework:** PyTorch, Keras, ODB, Kitura, Lucene, Jinja2
- **Language:** GRE - 333/340 + 5/6 (Verbal 163, Quantitative 170, Analytical Writing 5); TOEFL iBT 113/120 (Reading 30, Listening 29, Speaking 26, Writing 28)